Question ID: eae66bf9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-Text Connections	Medium

Question
Text 1

In 2021, a team led by Amir Siraj hypothesized that the Chicxulub impactor—the object that struck the Yucatán Peninsula sixty-six million years ago, precipitating the mass extinction of the dinosaurs—was likely a member of the class of long-period comets. As evidence, Siraj cited the carbonaceous chondritic composition of samples from the Chicxulub impact crater as well as of samples obtained from long-period comet Wild 2 in 2006.

Text 2

Although long-period comets contain carbonaceous chondrites, asteroids are similarly rich in these materials. Furthermore, some asteroids are rich in iridium, as Natalia Artemieva points out, whereas long-period comets are not. Given the prevalence of iridium at the crater and, more broadly, in geological layers deposited worldwide following the impact, Artemieva argues that an asteroid is a more plausible candidate for the Chicxulub impactor.

Based on the texts, how would Artemieva likely respond to Siraj's hypothesis, as presented in Text 1?

Answer
A. By insisting that it overestimates how representative Wild 2 is of long-period comets as a class

B. By arguing that it does not account for the amount of iridium found in geological layers dating to the Chicxulub impact

C. By praising it for connecting the composition of Chicxulub crater samples to the composition of certain asteroids

D. By concurring that carbonaceous chondrites are prevalent in soil samples from sites distant from the Chicxulub crater

Correct Answer: B

Rationale
Choice B is the best answer. Siraj's hypothesis is that the Chicxulub impactor was a long-period comet. But Artemieva points to the iridium found in the crater and in "geological layers that were deposited worldwide after the impact" as evidence that it was actually an asteroid, not a long-period comet.

Choice A is incorrect. We can't infer that this is how Artemieva would respond to Siraj's hypothesis. Text 2 never discusses whether Wild 2 is representative of long-period comets in general. Rather, Text 2 presents Artemieva's argument that the Chicxulub impactor was an asteroid, not a long-term comet. Choice C is incorrect. We can't infer that this is how Artemieva would respond to Siraj's hypothesis. Siraj's hypothesis doesn't make this connection: rather, Siraj hypothesizes that the Chicxulub impactor was a long-term comet. Choice D is incorrect. We can't infer that this is how Artemieva would respond to Siraj's hypothesis. "Soil samples from sites distant from the Chicxulub crater" is too vague. Only soil samples from sites that are connected to the impact in some way are involved in either hypothesis.